

Test of Mathematics for University Admission
Mock Set G Paper 1
 Applications of Mathematical Knowledge

Time allowed: **75 minutes**
 Questions: 20 multiple choice
 Calculators: not permitted
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Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Every question uses AS content only. The difficulty lies in how that content is used: parameters that change the case, conditions that quietly remove a solution, and answers that come from structure rather than from brute force.

[illegible]

Question 1

An equilateral triangle T_1 has side length 1. A circle C_1 is inscribed in T_1 . An equilateral triangle T_2 is then inscribed in C_1 , with its three vertices on C_1 ; a circle C_2 is inscribed in T_2 ; an equilateral triangle T_3 is inscribed in C_2 ; and the process continues without end, producing circles $C_1, C_2, C_3, \dots$. Find the total area of all the circles $C_1, C_2, C_3, \dots$.

- A $\frac{\pi}{12}$
- B $\frac{\pi}{10}$
- C $\frac{\pi}{9}$
- D $\frac{\pi}{8}$
- E $\frac{\pi}{6}$
- F $\frac{\pi}{3}$

Question 2

In the expansion of $(1+2x)^n$ in ascending powers of x , where n is a positive integer, there are three consecutive terms whose coefficients are in the ratio 1 : 4 : 10. If these are the terms in x^{r-1} , x^r and x^{r+1} , find $n+r$.

- A 7
- B 9
- C 10
- D 11
- E 13
- F 14

Question 3

In triangle ABC , $AB = 3$ and $AC = 5$. The median from A , that is the line segment from A to the midpoint M of BC , is perpendicular to the side AB . Find the length of BC .

- A $\sqrt{34}$
- B $2\sqrt{13}$
- C $\sqrt{58}$
- D 8
- E $2\sqrt{17}$
- F $\sqrt{73}$

Question 4

Find the sum of all the real solutions of the equation

$$\log_2(x+3) + \log_4((x-1)^2) = \log_2 3.$$

- A $\sqrt{7} - 1$
- B -2
- C 0
- D $\sqrt{7} + 1$
- E $\sqrt{7} - 3$
- F $-1 - \sqrt{7}$

Question 5

A circle passes through the points $(0, 0)$ and $(4, 0)$ and touches the line $y = x + 4$. There are two such circles. Find the sum of their radii.

- A $12\sqrt{2}$
- B $8\sqrt{2}$
- C $10\sqrt{2}$
- D 16
- E $6\sqrt{2}$
- F $4 + 8\sqrt{2}$

Question 6

Real numbers x and y satisfy $x + y = 4$. Find the minimum value of

$$(x^2 + 1)(y^2 + 1).$$

- A 9
- B 13
- C 16
- D 17
- E 20
- F 25

Question 7

Evaluate

$$\sum_{k=1}^{89} \frac{1}{1 + \tan k^\circ}.$$

- A 1
- B 44
- C 45
- D 89
- E $\frac{89}{4}$
- F $\frac{89}{2}$

Question 8

An open box is made from a rectangular sheet of card measuring 12 cm by 8 cm by cutting a square of side x cm from each corner and folding up the sides. Find the maximum possible volume of the box, in cm^3 .

- A 64
- B $\frac{1792}{27}$
- C $\frac{640 + 224\sqrt{7}}{27}$
- D $\frac{448 + 640\sqrt{7}}{27}$
- E 70
- F $\frac{640 + 448\sqrt{7}}{27}$

Question 9

The polynomial $p(x)$ has degree 4 and leading coefficient 1, and $p(1) = p(2) = p(3) = p(4) = 5$.

Find $\int_1^4 p(x) dx$.

- A 15
- B $\frac{141}{10}$
- C $\frac{159}{10}$
- D $\frac{27}{2}$
- E $\frac{33}{2}$
- F $\frac{147}{10}$

Question 10

Let $a > 0$. Consider the inequality

$$|x^2 - 2x - 3| \leq a(x + 1).$$

For which values of a is its complete solution set a single interval, with no isolated points?

- A $a > 0$
- B $a \geq 2$
- C $a > 4$
- D $a \geq 4$
- E $0 < a \leq 4$
- F no value of a

Question 11

Let $k > 0$. The finite region enclosed between the curve $y = |x^2 - k|$ and the line $y = k$ has area $\frac{64}{3}(\sqrt{2} - 1)$. Find k .

- A 4
- B 2
- C 8
- D $2\sqrt{2}$
- E 16
- F $\sqrt{2}$

Question 12

For each real k , let $N(k)$ be the number of real solutions of the equation

$$|x^2 - 4|x| + 3| = k.$$

Find $N(0) + N(1) + N(2) + N(3) + N(4)$.

- A 15
- B 17
- C 19
- D 20
- E 21
- F 23

Question 13

How many real solutions does the equation

$$2^x + 3^x = 4^x + 1$$

have?

- A 0
- B 1
- C 2
- D 3
- E 4
- F infinitely many

Question 14

For each real k , let $N(k)$ be the number of real solutions of

$$\left| \left| |x| - 3 \right| - 2 \right| - 1 = k.$$

Find $N(0) + N\left(\frac{1}{2}\right) + N(1) + N(2)$.

- A 23
- B 25
- C 27
- D 29
- E 31
- F 35

Question 15

The circle C_1 has centre $(0, 0)$ and radius 1, and the circle C_2 has centre $(4, 0)$ and radius 2. There are exactly four lines that are tangent to both circles. Find the sum of the squares of their gradients.

- A $\frac{142}{105}$
- B $\frac{284}{105}$
- C $\frac{2}{15}$
- D $\frac{18}{7}$
- E $\frac{16}{7}$
- F $\frac{64}{21}$

Question 16

In the quadrilateral $ABCD$, $AB = BC = 1$ and $\angle ABC = 90^\circ$. The point D lies on the opposite side of the line AC from B , with $\angle ADC = 45^\circ$. Find the greatest possible length of BD .

- A $\sqrt{2}$
- B 2
- C $\sqrt{5}$
- D $1 + \sqrt{2}$
- E $1 + \sqrt{3}$
- F 3

Question 17

Every tangent to the curve $y = x^3 - 6x^2 + 9x$ that passes through the origin is drawn. Find the total area of all the finite regions enclosed between the curve and these tangents, for $x \geq 0$.

- A $\frac{27}{4}$
- B 108
- C $\frac{405}{4}$
- D $\frac{459}{4}$
- E $\frac{513}{4}$
- F $\frac{243}{2}$

Question 18

For each real k with $-16 < k < 16$, the equation $x^3 - 12x + k = 0$ has three distinct real roots $\alpha < \beta < \gamma$. As k varies over this interval, which of the following is the set of all possible values of $\gamma - \alpha$?

- A $6 < \gamma - \alpha < 8$
- B $0 < \gamma - \alpha \leq 4\sqrt{3}$
- C $6 < \gamma - \alpha \leq 4\sqrt{3}$
- D $4\sqrt{3} \leq \gamma - \alpha < 8$
- E $6 \leq \gamma - \alpha \leq 4\sqrt{3}$
- F $\gamma - \alpha = 4\sqrt{3}$ for every k

Question 19

Real numbers x , y and z satisfy $x + y + z = 3$ and $x^2 + y^2 + z^2 = 5$. Find the greatest possible value of xyz .

- A $\frac{1}{3}$
- B $\frac{\sqrt{3}}{9}$
- C $\frac{4}{9}$
- D $\frac{2\sqrt{3}}{9}$
- E 1
- F $\frac{2}{3}$

Question 20

Let $f(x) = x^2 + c$. Find the complete set of values of c for which the equation $f(f(x)) = x$ has exactly two distinct real solutions.

- A** $-\frac{3}{4} < c < \frac{1}{4}$
- B** $-\frac{3}{4} \leq c < \frac{1}{4}$
- C** $-\frac{3}{4} \leq c \leq \frac{1}{4}$
- D** $c < \frac{1}{4}$
- E** $-1 < c < \frac{1}{4}$
- F** $c \leq -\frac{3}{4}$

END OF PAPER